

Crisphine M. Ngari

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Research Interests

Analog and mixed-signal circuit design • Digital VLSI • Wireless communication • Hardware and IoT security • Machine learning for network intrusion detection

Education

Southern Illinois University Carbondale

Aug 2025 – Present

M.S., Electrical and Computer Engineering GPA: 3.89 / 4.0

Carbondale, IL

- Advisor: Prof. Ning Weng

GITAM School of Technology

Sep 2020 – May 2024

B.Tech., Electronics and Communication Engineering First Class, GPA: 8.24 / 10

Bangalore, India

- Thesis: *Wireless Localization on Constrained Hardware*
- Study in India Merit-Based Scholarship (2020–2024)

Publications

Under Review

- **N. C. Macharia**, N. Weng, N. Yang. “Evaluation of Concept-Based Interpretability and Neuro-Symbolic Out-of-Distribution Detection.” *Under review*, 2025.

Conference Papers

- **N. C. Macharia** et al. “Wireless Localization on Constrained Hardware.” In *Proc. IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, 2024.

Research & Projects

Interpretable IoT Intrusion Detection with Open-Set Robustness

2025 – Present

Southern Illinois University Carbondale

- Jointly evaluated Concept Bottleneck Models (CBMs) and Neuro-Symbolic NIDS (NeSy-NIDS) on IoT traffic classification using CTU-IoT-23 and CIC-IoT-2023 datasets.
- Both architectures use Mahalanobis distance for open-set scoring; neither incurs classification cost (F1 within ± 0.0015 of an unconstrained MLP).
- NeSy-NIDS achieved OOD AUROC of 0.909 on CTU-IoT-23.

Sneak Path Mitigation in a Crossbar-Based AI Accelerator

2024 – 2025

Southern Illinois University Carbondale

- Investigated the sneak path problem in resistive memristor crossbar arrays for in-memory neural network inference.
- Designed a 1D1R (diode + memristor) cell in Cadence Virtuoso, reducing read error from 3,430% to under 10%, enabling practical multiply-accumulate operations for edge AI.

Full-Custom CMOS XOR Gate Layout and Verification

2024 – 2025

Southern Illinois University Carbondale

- Designed a 2-input XOR gate from schematic to physical layout in 45 nm GPDK45 using a complementary pass-transistor topology in Cadence Virtuoso.
- Achieved zero DRC and LVS violations, demonstrating the complete RTL-to-GDS flow.

Wireless Localization on Constrained Hardware (B.Tech Thesis)

Jul 2023 – Apr 2024

GITAM School of Technology

- Designed a real-time Wi-Fi fingerprint localization framework for IoT nodes, achieving 99% accuracy with an average response time of 3 seconds.
- Addressed the self-localization challenge for resource-limited devices in asset tracking and indoor navigation applications.
- Published at IEEE CONECCT 2024.

Teaching & Academic Service

Graduate Teaching Assistant — Southern Illinois University Carbondale *Aug 2025 – Present*

- ECE 513: Digital VLSI Design (Spring 2025) — lab instruction, grading, office hours
- ECE 433: Networking (Fall 2025) — assignment grading and student support

Professional Experience

Afrotel Group Telecommunication Engineering | Network Engineer *Jun 2024 – Jul 2025*

- Installed and maintained LTE, 5G RAN, microwave, and FTTX fiber optic infrastructure.
- Conducted OTDR, CD/PMD, and power meter tests; performed splice quality validation and point-of-break analysis.
- Troubleshoot network faults to prevent downtime; managed MPLS data center connectivity.

EduSkills Foundation | Embedded Systems Intern *May 2023 – Jul 2023*

- Developed and tested embedded hardware/software solutions integrating sensors, actuators, and communication modules.
- Performed circuit-level troubleshooting on resistive, capacitive, and diode components.

South Eastern Kenya University | ICT Technician *Sep 2019 – Feb 2021*

- Managed network infrastructure, computer labs, and CCTV systems; provided ICT training for students and staff.

Technical Skills

EDA & Design Cadence Virtuoso, SPICE simulation, 45 nm GPDK45

HDL VHDL, Verilog, FPGA/CPLD

Programming Python, MATLAB, Simulink

Networking LTE/5G, FTTX, MPLS, RSSI/Wi-Fi fingerprinting

Hardware Arduino, Raspberry Pi, NodeMCU

Awards & Scholarships

- Study in India Merit-Based Scholarship (2020–2024)

Languages

English (bilingual proficiency) | Swahili (native speaker)